



Control of nanopositioning and nanomeasuring machines with a modular FPGA based data processing system

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ABSTRACT

For precise length measuring techniques, highly dynamic positioning stages are used to measure and manipulate objects with a resolution in subnanometer range. The position and the orientation of these systems have to be controlled permanently to compensate external disturbances such as sound waves, thermal expansion of the mechanical components, and ground motion. Because of new applications in the area of microsensing devices the dynamic or path accuracy is in addition to the stationary accuracy a critical feature. To improve the dynamic behavior of these machines, an efficient friction compensation as well as a high performance signal- and data processing system is necessary. This work presents a realization of a control system with friction compensation for a nanopositioning and nanomeasuring machine based on FPGA technology. Particular attention was paid to the obtainable sampling rate and its impact on the attainable path accuracy.

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1. Introduction

The miniaturization in several high-tech industries and in laboratory has increased the need for dimensional micro- and nanometrology. There are several applications, where objects have outer dimensions in the millimeter range, while the structures and features on this objects reach atomic dimensions. For this reason the Collaborative Research Centre 622 (SFB 622) at the Ilmenau University of Technology has been working on the development of Nanopositioning and Nanomeasuring Machines (NPM). The current generation of these high-precision measuring machines has a positioning volume in the centimeter range but a resolution and reproducibility in the nanometer range.

With the ongoing development of microsensing devices it is now possible to realize a true 3D coordinate metrology with nanometer precision. Due to their shape, sensing devices are able to perform sidewall measurements or detect under-cuts [1]. If the sensing device is placed within a deepening, it is at risk of being damaged in case of an unfavorable movement. This scenario shows that only a stationary accuracy in the nanometer range is insufficient. For this new challenge it is essential to reach an extremely high path accuracy. Hence, a significant improvement regarding the dynamic behavior of the NPM is inevitable.

Previous analyses have shown [2–4] that this goal can only be reached by estimating and compensating the friction force, which is caused by the used ball bearings.

In literature various friction models can be found, for example single state models like the elasto-plastic [5] or multi state models like the generalized maxwell slip model [6]. For the control of an NPM model based, as well as non-model based approaches were examined.

In addition to an effective friction compensation, the reachable sampling rate of the signal and data processing is of vital importance for the velocity and accuracy characteristics of the overall system. This can be seen by a rough calculation. In case of movements with 1 mm/s and a cyclic time of 100 µs the spatial step size is 100 nm.

This contribution discusses the realization of a control system of a NPM. It was designed hardware independent and the final realization was done on FPGA¹ technology. For friction compensation a Kalman filter based estimation is used. Particular attention lies on the obtainable sampling rate and its impact on the attainable path accuracy.

After presenting the addressed NPM in Section 2 the control system is described in Section 3. Furthermore, the realization of the control system on modular FPGA-based signal- and data processing is introduced. It is shown by experimental results (Sec-

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¹ FPGA – field programmable gate array.

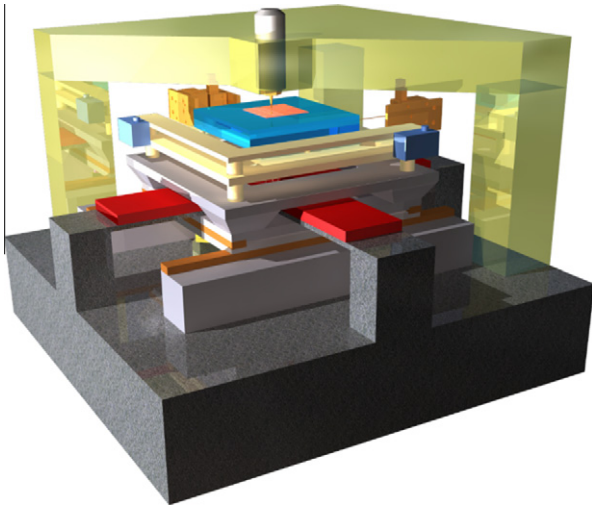


Fig. 1. Draft of a NPM with a moving range of $200 \times 200 \times 25 \text{ mm}^3$.

tion 5) that a drastic increase of the sampling rate improves the accuracy significantly.

2. Laboratory set-up of the positioning system

The experimental set-up is a prototype of a NPM with a planar moving range of $200 \times 200 \text{ mm}^2$ (see Fig. 1).

2.1. Mechanical set-up

The overall system is placed on a vibration-damped fundament to minimize the impact of vibrations. Each length and angle measuring system as well as the probe system is attached to a metrological frame of Zerodur (see Fig. 2), a glass–ceramic with a thermal expansion coefficient of approximately zero. Thereby, thermal influences towards the measuring system are minimized. The corner mirror, which also consists of Zerodur, is fixed on a three-axis positioning stage. The two planar moving axes are supported by two linear V-grooved high precision roller bearings of the company Schneeberger [7]. The y-axis is arranged on the x-axis. The x- as well as the y-axis are driven by planar motors and the primary part of these engines are mounted on the fundament of the NPM (red² in Fig. 1). The position of the corner mirror is measured by stabilized HeNe-laser interferometers of the company SIOS Meßtechnik GmbH [8]. These interferometers reach a position resolution of 0.08 nm.

2.2. Power system

The x-axis as well as the y-axis is driven by two asynchronous linear motors. The engines are powered by internal developed analog amplifiers. Hence, the necessary current can be provided with the required precision. The commutation of the linear motors depends on the position and is realized software-based by the real-time system. The analog amplifiers are controlled via a 16-bit digital/analog converter with voltage in range of $\pm 10 \text{ V}$. Due to an integrated current control, the power electronics allocates an output current in range of $\pm 3 \text{ A}$, which is proportional to the input voltage.

The power amplifiers were activated with stepped signals in order to develop a model of the drive system and to test the manufacturer's specifications. Based on the measured currents a system model was established. This model consists of a first-order

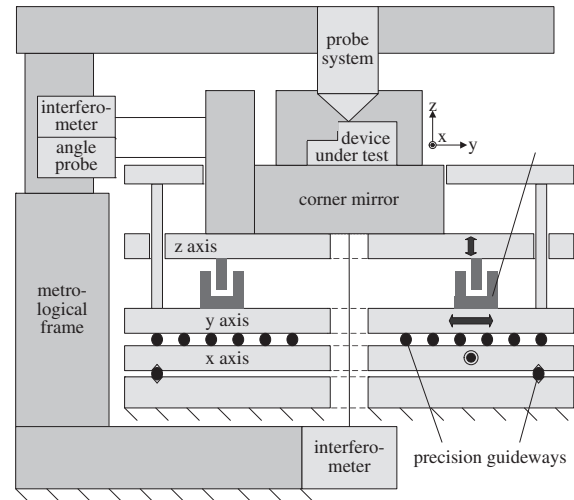


Fig. 2. Basic structure of a NPM.

delay and a limitation of the current increase in the form of a rate limit. The identified parameters are $T_1 = 135 \mu\text{s}$ for the first-order delay and $12 \times 10^3 \text{ A/s}$ for the rate limit. Fig. 3 depicts the result of this model identification. Due to this model the 3 dB cutoff frequency of the power system can be detected as 1180 Hz.

2.3. Signal- and data processing

Within the experimental set-up, PXI-systems from National Instruments are used to realize the real-time relevant signal- and data-processing. The Signal- and Data Processing Unit (SDPU) is a composition of three PXI-chassis with controllers and miscellaneous FPGA-modules. The overall system, including the SDPU, is depicted in Fig. 4.

The division of the SDPU into three sub-systems, each realized by a PXI-system, is based on functional and logical aspects: one sub-system is used for data acquisition (Data Acquisition System – DAS), one sub-system is used for the higher control of the measurement tasks (Sequence Control System – SCS) and finally the third sub-system realizes the control system and directly controls the drives (Controller System – CS). Regarding the analysis of the path accuracy and the control's efficiency, only the part of the DAS, which deals with the position determination, is relevant. A detailed description of the used positioning system can be found in [9]. The system for the position evaluation processes 18 analog signals (position and angle signals) with $666\frac{2}{3} \text{ kHz}$. These signals result in raw measurement data values which are further processed with a frequency of $83\frac{1}{3} \text{ kHz}$. Complex calculations for on-line value correction with floating-point arithmetic (single and double precision) do not violate the high real-time requirements (measurement cycle time of $12 \mu\text{s}$). Therefore, it is possible to generate position data for the position control with a maximum frequency of $83\frac{1}{3} \text{ kHz}$. This is the initial point for the following studies. Consecutively, the control system and its realization are further examined.

3. Control system

The control system of the NPM has to mainly fulfill two tasks. Firstly, following the reference trajectory without tracking errors, what demands a precise friction compensation (see [3]), and secondly to oppress external disturbances like vibrations and sound waves. Fig. 5 shows the schematics of the used control system. It

² For interpretation of color in Fig. 1, the reader is referred to the web version of this article.

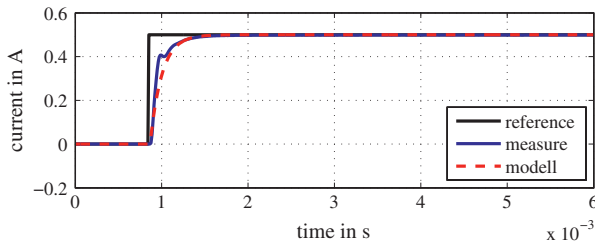


Fig. 3. Identification of the drive system.

is able to speed up the dynamic behavior of the NPM and to improve the robustness against external disturbances.

3.1. Trajectory generation

The trajectory generation algorithm calculates the dynamic set-points. To prevent vibrations of the mechanical system induced by steps in the reference position and its derivatives, the calculated set-points are continuously differentiable to the derivative of the jerk. fourth-order trajectory generation. The planning algorithm is an extension of the algorithm proposed by Lambrechts et al. [10,11]. The planned trajectory is based on the construction of the djerk profile. This djerk trajectory is composed of piecewise defined quadratic functions. The trajectory can be analytically integrated several times in order to obtain a continuously differentiable jerk, acceleration, velocity and position trajectory. A further feature of the algorithm is the ability to plan motions with random initial and final velocities. This provides the ability to realize continuous trajectory planning over an arbitrary number of connected path segments. In addition, kinematic restrictions of the mechanical systems, like maximum acceleration and maximum jerk, are considered explicitly. Thereby, it is ensured that the demanded target positions can be realized by the NPM. A detailed description of the used algorithm can be found in [12].

3.2. Feedback controller

The feedback controllers used in the NPM are time discrete PID controllers. The actuating variable is calculated based on the position- and velocity-error. Especially for small deviations the friction in the bearings behaves like a spring. This leads to enhanced position errors. After overcoming sticking friction, this effect is not visible anymore. Hence, a nonlinear modification of the integrator amplification (see Fig. 6), which is proposed by Hausotte et al. [13], improves fine positioning. The resulting feedback control law is

$$F_{pid} = K_p e + K_i(e) \int e dt + K_d \dot{e} \quad (1)$$

with $e(t) = x_{ref}(t) - x(t)$ and $K_i(e) = k_i \cdot g(e)$.

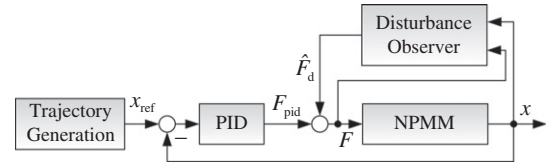


Fig. 5. Control scheme of one axis of the NPM.

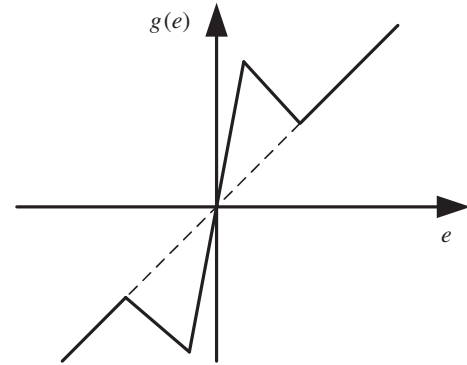


Fig. 6. Nonlinear modification of the integrator amplification.

3.3. Friction observer

In addition, a friction compensation based on a Kalman filter is used to reach the demanded accuracy. The utilized approach was first presented by Ramasubramanian and Ray [14]. The following equation:

$$F(t) = m \cdot \ddot{x} + F_f(t) \quad (2)$$

describes the underlying continuous system model. The driven force $F(t)$ accelerates the mass m and helps to overcome the friction force $F_f(t)$. In this force, besides friction also other disturbances are concentrated which are caused by dust, wiring, mechanical tolerances and so on. The dynamics of the mechanical system and the precise position measurements are used to calculate this unknown friction force $F_f(t)$. In [15] a stability analysis and performance investigations of this approach were provided. In state space notation the unknown force $F_f(t)$ is modeled as a second order random walk (Eq. (3)), which means that second derivative of this force is supposed to be zero.

$$\begin{bmatrix} \dot{x}(t) \\ \dot{x}(t) \\ \dot{F}_f(t) \\ \dot{F}_f(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -1/m & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} x(t) \\ \dot{x}(t) \\ F_f(t) \\ \dot{F}_f(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1/m \\ 0 \\ 0 \end{bmatrix} \cdot F(t) \quad (3)$$

Transforming the equation into discrete time domain leads to the state and measurement equations of the Kalman filter

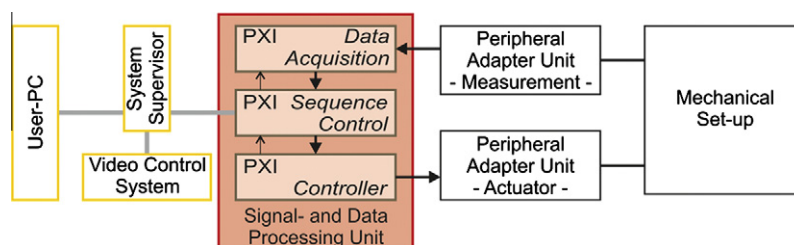


Fig. 4. The signal- and data processing unit as part of the overall system architecture.

$$\mathbf{x}_k = \Phi \mathbf{x}_{k-1} + \Gamma \mathbf{u}_k + \mathbf{v}_k \quad (4)$$

$$\mathbf{z}_k = \mathbf{H} \mathbf{x}_k + \mathbf{w}_k \quad (5)$$

with $\mathbf{x}_k = [x_k \dot{x}_k F_{fk} \dot{F}_{fk}]^T$ and $\mathbf{u}_k = F_k$. The state transition matrix is

$$\Phi = \begin{bmatrix} 1 & T & -\frac{T^2}{2m} & -\frac{T^3}{6m} \\ 0 & 1 & -\frac{T}{m} & -\frac{T^2}{2m} \\ 0 & 0 & 1 & T \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

and the control effect matrix is $\Gamma = [\frac{T^2}{2m} \frac{T}{m} 0 0]^T$. The process noise is described by $\mathbf{v}_k$ and $\mathbf{w}_k$ is the measurement noise and $\mathbf{H}$ is the output matrix.

The Kalman filter estimates the state vector in each time step according to the following equation:

$$\hat{\mathbf{x}}_k = \Phi \hat{\mathbf{x}}_{k-1} + \Gamma \mathbf{u}_k + \mathbf{K}_k (\mathbf{z}_k - \mathbf{H} \Phi \hat{\mathbf{x}}_{k-1} - \mathbf{H} \Gamma \mathbf{u}_k) \quad (6)$$

$\mathbf{K}_k$ are the Kalman gains. The estimated force $\hat{F}_{fk}$ is part of the estimated state vector. This force reduces the influence of the nonlinear friction, which is introduced into the system by the used ball bearings as well as further external disturbances like vibrations or soundwaves. This approach was already proved for NPMs [2].

4. Realization of the control system

In the following section, the implementation of the control system, which is described above, is presented. Firstly, the hardware components, the emerging architecture as well as the specific software components are described. Afterwards, the resource and the timing properties are discussed.

4.1. Hardware platform and architecture of the control system

The PXI-systems from National Instruments are used due to compatibility to the position measuring system introduced above. The modular system consists of a PXI-Chassis (PXI-1042Q) with a PC-based Realtime-Controller (PXI-8108-RT) and two FPGA-Modules (PXI-7853R). These FPGA-modules have eight analog differential inputs (750 kHz and 16 bit resolution), eight analog single-ended outputs (750 kHz and 16 bit resolution) and 96 digital in-/outputs (40 MHz). The built-in FPGA is a Virtex-5-LX-85. 11 analog output signals are used to control the drives. Four analog signals are needed to control the x- and y-drives respectively. The remaining three output signals are intended to control the z-drives. They are not relevant for the experiments presented in this work. Hence, two FPGA-modules are utilized to control all three movement-axes.

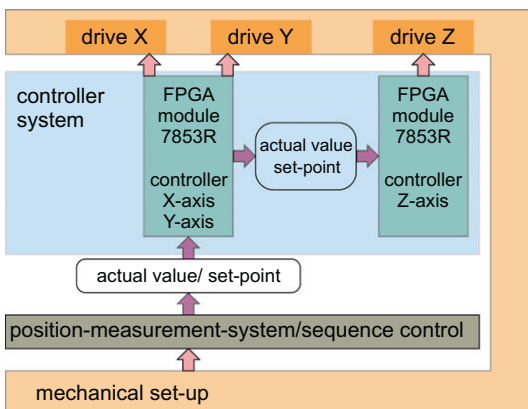


Fig. 7. Architecture of the control system.

Fig. 7 shows the basic architecture of the control system in relation to the overall system. In addition, the data flow is visualized. The purple arrows mark the analog in- and output signals. They interface the mechanical structure and the signal and data processing unit. The pink arrows represent the data transfer, which are exchanged within the real-time system. In the position measuring system the actual position is determined and the position set-points are generated. Subsequently, these data (actual position and set-points) are transferred to the control system, where they are being further processed. The data transmission is carried out via the digital interfaces of the FPGA-modules due to the high real-time requirements. In contrast, the communication via the PCI-backplane would implicate too high time delays. Consequently, the calculation must be carried out close to the process. In application, this implies that the control must be calculated on the FPGAs.

4.2. Implementation in LabVIEW

Some characteristics have to be observed concerning the control within the FPGA-module. The self-developed digital communication protocol meets the high performance requirements. The data transfer between the two FPGA-modules can be realized without any problems. The most important challenge is the implementation of the control algorithms, which consists of 394 floating point additions and floating point subtractions, 327 floating point multiplications and three floating point divisions. As described above, the algorithm development was done without consideration of a concrete hardware platform. In principle, a floating point data type with double precision was chosen. This leads to challenges in the realization of the algorithms on the FPGA, because of its limitation in offering only integer and fixed-point computation as well as limited space capabilities.

To ensure the necessary precision, the algorithms need to be implemented with floating point arithmetic on the FPGA. Because of special generatable operation blocks, floating point arithmetic can be utilized within a FPGA. A direct realization on a FPGA-module with LabVIEW is not possible due to the limited resources and the exceedence of the requested maximum cycle time of 12 μ s of some solutions.

To solve this problem, a proprietary synthetic processor was designed and developed [16,17]. This processor includes an adaptable and extendable LiSARD-calculation engine (LabVIEW integrated Softcore Architecture for Reconfigurable Devices) as LabVIEW-VI, and VIs for software development, configuration and test of programs. With the help of this softcore, complex algorithms can be executed locally on a FPGA-module. The performance of this component is shown in Table 1. The presented performance properties rely on the comparison in [16], which uses the Kalman filter described above.

The presented control system was realized using this softcore. The control algorithms for the x and the y axis were realized on one FPGA-module according to the architecture pictured above (Fig. 7).

4.3. Time behavior and usage of resources

The timing properties of the control can be expressed depending on the time gap between the actual values and the set-points. This time corresponds to the calculation time in the context of the sequence control and the trajectory generator. This leads to the following equation: $Duration = 7 \mu s + \max(1.6 \mu s; \Delta(t_{set-point} - t_{actual}))$. Hence, the minimum duration is 8.6 μ s. This corresponds to a maximum control rate of approximately 115 kHz. This exceeds the highest possible rate, which is limited by the occurrence of the measurement data. In the experiment described below, the time

Table 1

Performance comparison of different architectures (Kalman filter).

Architecture type	Computing time (without communication)	Computing time (incl. communication)
RT-Controller (NI PXI-8108RT)	0.18–2.3 μ s	30–33 μ s
DSP (TI C6713b)	6.4 μ s	8.0 μ s
LiSARD (NI PXI-7853R)	1.8 μ s	2.0 μ s

needed for the calculation of the sinusoidal movement is lower than 1.6 μ s due to the additional precalculation for the distribution over the coils of the motor. The consumption of resources for a FPGA-module of type 7853R is: 72.9% Total Slices, 52.1% Slice Registers, 44.9% Slice LUTs, 27.1% DSP48s, 16.7% Block RAMs.

5. Subject of the investigation

In this chapter two main questions were analyzed with several experiments. On the one hand the benefit of the designed disturbance observer especially at high dynamic motions is shown. On the other hand the influence of the sampling rate with respect to the performance of the control system is examined.

5.1. Test set-up

Caused by the time consuming and complex communication and computation within the sequence control system the control frequency is limited to 8 $\frac{1}{3}$ kHz. To enhance this limitation in contrast to the architecture of the original signal- and data processing unit, in this investigation the architecture is altered. Transferring a reduced sequence control and trajectory generation algorithm to an additional FPGA-module for experimental analysis enables to speed-up the control frequency up to 83 $\frac{1}{3}$ kHz and results in the system architecture shown in Fig. 8.

5.2. Test procedure

In order to check the functionality of the developed and implemented control system one axis of the NPMM was excited by a reference trajectory in the form of a sinusoidal movement with amplitude of 100 μ m and a frequency of 0.2 Hz (see Fig. 9). Due

to the fact, that all friction regimes were cyclically driven through, the capability of the proposed friction cancellation can be examined very well.

The performance of the control system was examined with and without friction compensation. The parameters of the controller were heuristically determined and are certainly not optimal. Therefore, the results do not allow statements about the absolute precision of the NPMM because this is not the focus of this contribution. The so-called Normalized Root Mean Square Error (NRMSE) (see Eq. (7)) serves as criterion for performance evaluation. Position datasets with duration of 15 s were utilized to calculate the NRMSE, which is defined as follows:

$$\text{NRMSE} = \sqrt{\frac{1}{\sigma_{x_{\text{ref}}}^2 \cdot N} \sum_{k=1}^N (x_{\text{ref}}(k) - x(k))^2} \cdot 100\% \quad (7)$$

with $\sigma_{x_{\text{ref}}}^2$ is the variance of the reference position, N the number of samples, x_{ref} the reference position and x the measured position. It has to be mentioned that all signals used for analysis and calculation of the NRMSE have been sampled with the same frequency. So the number of samples is always the same and the NRMSE is not falsified.

5.3. Performance according the control approach

For example Fig. 10 shows an experimental data-set composed of the reference, actual position trajectory (a) and the related position error (b). The position is controlled without a friction cancellation and it can be clearly seen, that the feedback controller is not able to follow the reference trajectory satisfyingly. The NRMSE is 17.5% and a significant phase shift is observable. For comparison Fig. 11 shows the behavior of the controlled system under operation of the proposed friction compensation. If the Kalman filter is

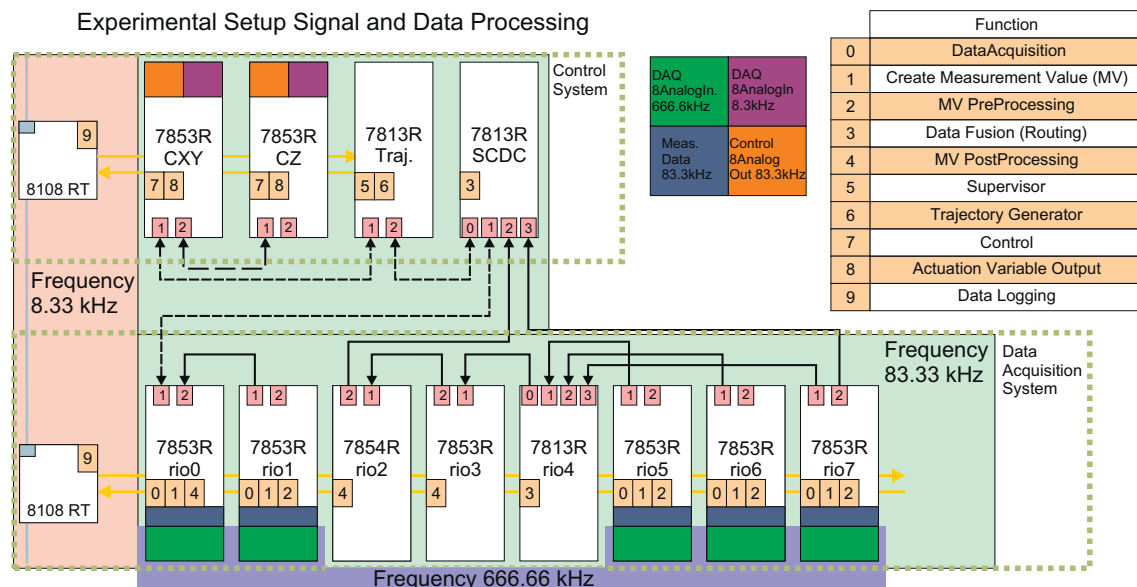


Fig. 8. Altered architecture of the signal- and data processing unit to enable a speed-up of the control frequency.

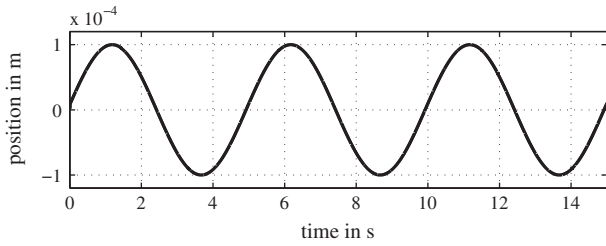


Fig. 9. Sinusoidal reference trajectory.

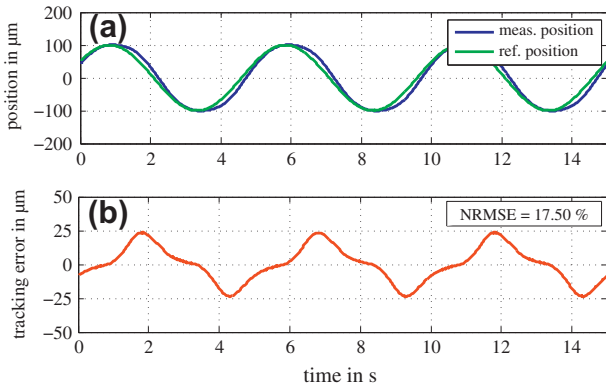


Fig. 10. Position reference versus actual position (a) and tracking error (b) without friction compensation.

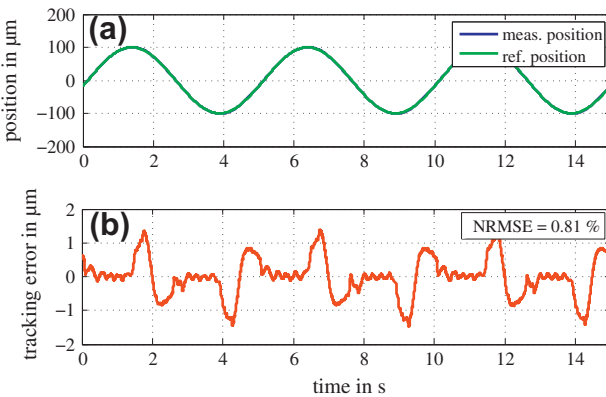


Fig. 11. Position reference versus actual position (a) and tracking error (b) with friction compensation.

utilized the system is able to follow the trajectory (see Fig. 11b) satisfactorily. As easily can be seen, the disturbance observer is able to reduce the amplitude of the tracking error nearly by factor of 18. The NRMSE is also minimized to 0.81%. These experiments show clearly the impact of the proposed friction cancellation approach to the performance of the control system.

5.4. Impact of the control frequency on the controller performance

In the first experiment, only the well tuned PID controller is used for position control. The reached positioning errors are shown in Fig. 12 depending on the control frequency. No clear tendency is recognizable. The positioning error is the lowest at 83.3 kHz but increases between 15 kHz and 40 kHz without apparent reason. Obviously, high clock speeds is not able to reduce the control error significantly with this configuration.

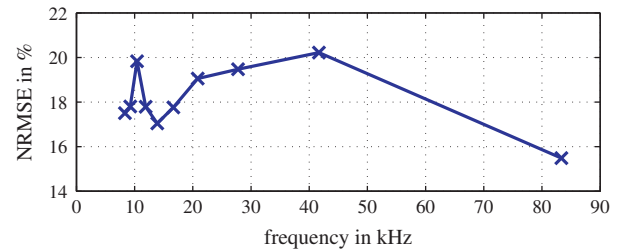


Fig. 12. Relative tracking error versus control frequency (PID controller).

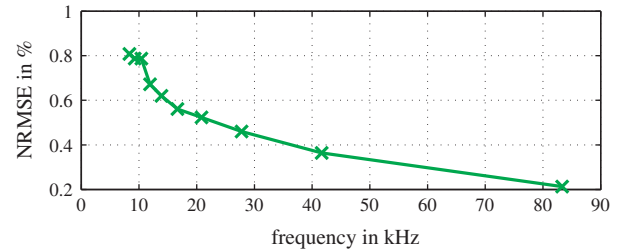


Fig. 13. Relative tracking error versus control frequency (PID controller with Kalman filter).

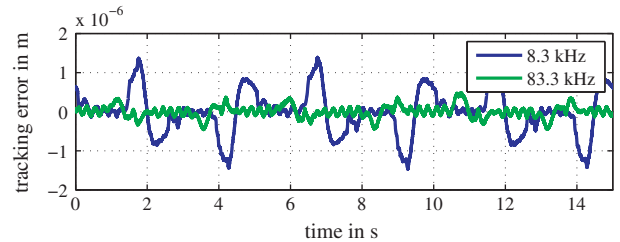


Fig. 14. Tracking error versus time for different control frequencies by using Kalman filter, $NRMSE(8.3 \text{ kHz}) = 0.81\%$, $NRMSE(83.3 \text{ kHz}) = 0.21\%$.

5.5. Results PID controller with Kalman filter

In the second experiment, a Kalman filter is used for disturbance compensation in addition to the PID controller. The reached tracking errors are depicted in Fig. 13. It can be seen clearly that all values are smaller than in Fig. 12. The maximum error is reduced from 20% to 0.8%. Apparently, there is also a significant improvement according to an increasing clock speed. The error decreases from 0.8% to 0.2%. The decay of the control error has approximately exponential character. This leads to the assumption that no arbitrary improvement is possible. Nevertheless, it is remarkable that the tracking error decreases with increasing clock speed although the frequency is many times greater than the limiting frequency. The direct comparison of the tracking error by using Kalman Filter for the lowest and the highest frequency is depicted in Fig. 14.

The presented results show the effectiveness of the proposed friction compensation scheme using a Kalman Filter. Furthermore the experiments demonstrates, that the Kalman filter estimates a more precise friction force by using a higher sampling rate. This can be explained by the functionality of the observer. Because of the second order random walk model, the friction force is almost completely calculated in the correction step of the Kalman filter. The often the correction is calculated, the better is the estimation of the friction force.

6. Conclusion

In this work, the hardware independent model-based design of a control algorithm including a disturbance observer for friction control in high precision coordinate-measuring machines is presented. This complex position control system of a NPMM and its implementation on a modular FPGA based data processing system is further reviewed. It was shown that even complex floating point algorithms can be realized within an FPGA-based control system. The presented control concept was realized within the FPGA-modules by using an especially developed softcore. A realization on an FPGA has the great advantage of a computation very close to the controlled process so very less communication costs occur. Based on the overall FPGA-based realization of the complete signal- and data processing unit it is in principle possible to speed-up the system frequency.

The main focus in the presented experiments lies on the analysis of the benefit of the designed disturbance observer and the influence of the sampling rate. Particular attention was paid to the performance of the data processing system. The reached performance parameters allow to experimentally increasing the clock speed of the control system up to $83\frac{1}{3}$ kHz which is derived of the data acquisition system. The tracking accuracy was examined by a sinusoidal movement along one axis. Although the motors have a limited frequency of 1.2 kHz, a significant reduction of the control error at high clock speeds can be achieved by disturbance compensation with a Kalman filter. Especially the disturbance observer profits from the computational power and the achievable higher clock rates. This combination leads to a further reduction of the control error and increases the quality of the NPMM. The results motivate to fully transfer the real-time depending computation onto the FPGA-modules to achieve high precision performance parameter.

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